Geospatial Analysis Report: National Trends in India's Sex Ratio

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1. Introduction: Defining Gender Equity Through Spatial Data

This report provides a macro-level analysis of India's sex ratio, utilizing geospatial visualization to identify patterns of gender equity across the country. By moving beyond tabular data, this spatial approach reveals clear regional clusters that are vital for public health analytics and social policy planning.

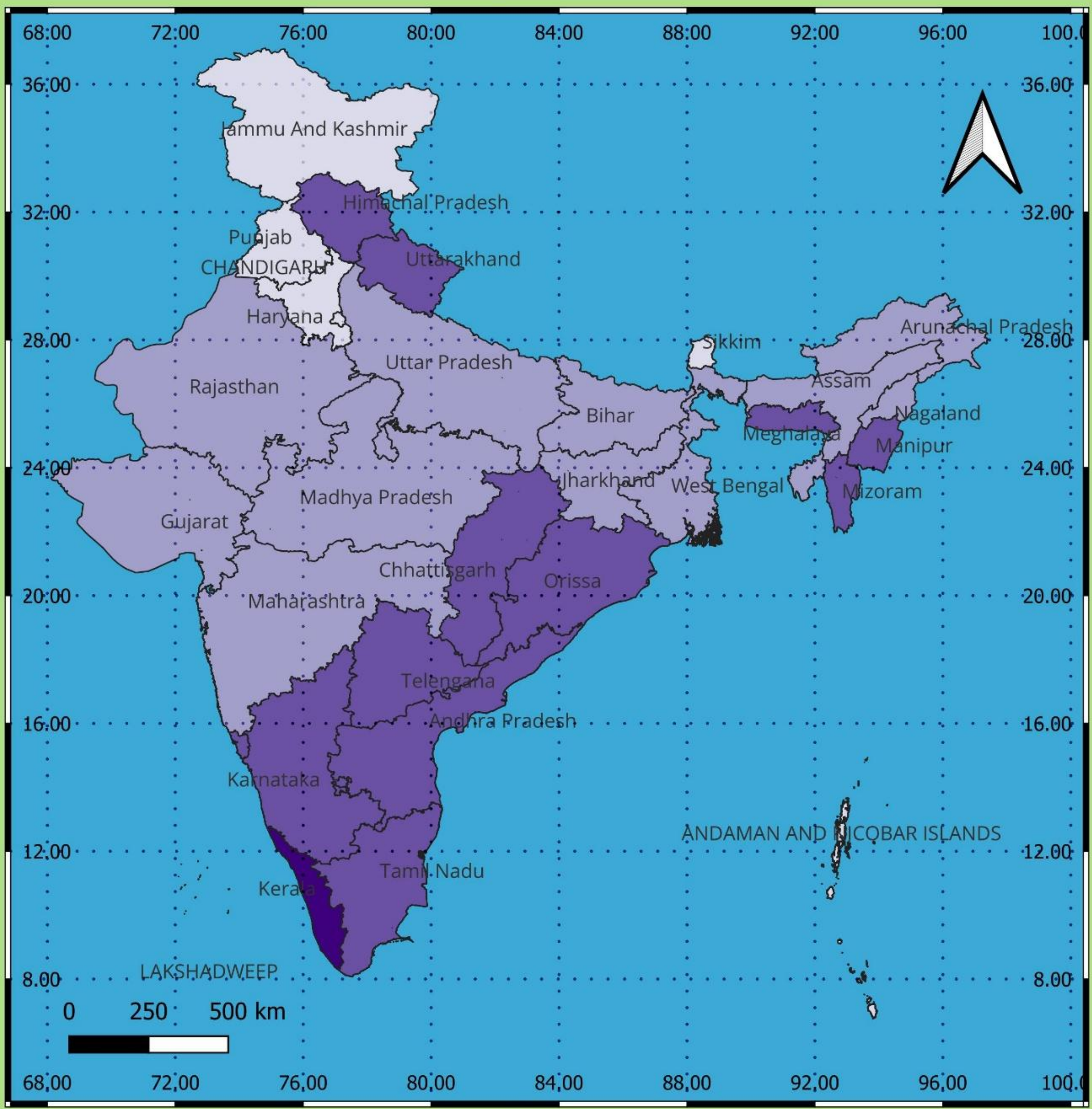
2. Methodology: Data Integration and Choropleth Mapping

The analysis was performed using **QGIS**, integrating Census of India datasets with national state boundaries.

- **Visualization Technique**: A choropleth map was engineered using five distinct data classes, ranging from 774 to 1084 females per 1000 males.
- **Data Classification**: Higher ratios are represented by deep purple tones, while lighter shades indicate lower ratios and higher gender imbalance.

3. Key Findings: The North-South Divide

SEX RATIO IN INDIA



The map shows clear regional variation in India's sex ratio. Southern and eastern states such as Kerala (1084), Tamil Nadu (996), and Odisha (979) record higher ratios, indicating a more balanced gender composition. In contrast, northern and northwestern states like Haryana (879), Delhi (868), and Punjab (895) show lower ratios, reflecting persistent gender imbalance. Overall, the map highlights India's north–south divide in gender equity, with southern regions performing significantly better.

India_SexRatio
- 774 - 836
- 836 - 898
- 898 - 960
- 960 - 1022
- 1022 - 1084

The most significant insight derived from this map is the stark **North-South divide** in gender composition.

- **Southern Strength**: The southern regions of India perform significantly better in gender equity metrics.
- **Northern Imbalance**: Northern and northwestern states exhibit persistent and severe gender imbalances.

4. Detailed Observations: Regional Benchmarks and Disparities

High-Performance States (Southern and Eastern)

These regions serve as the national benchmark for balanced gender composition:

- **Kerala**: Records the highest ratio at **1084**, indicating a female-majority population.
- **Tamil Nadu**: Shows a strong ratio of **996**.
- **Odisha**: Records a balanced ratio of **979**.

Low-Performance States (Northern and Northwestern)

These areas show clear regional variation where ratios fall significantly below the national average:

- **Haryana**: One of the lowest recorded ratios at **879**.
- **Delhi**: Shows severe imbalance at **868**.
- **Punjab**: Records a low ratio of **895**.

5. Strategic Implications for Analytics and Policy

- **Targeted Resource Allocation**: For healthcare and social analysts, this map pinpoints exact geographies where "Beti Bachao Beti Padhao" and similar gender-equality initiatives require the most intensive intervention.
- **Sociological Clustering**: The clustering of low ratios in the Northwest suggests that gender imbalance is not random but tied to regional socio-economic and cultural factors that require localized policy solutions.

- **Benchmarking Success**: The high ratios in Kerala and Tamil Nadu provide a spatial "success model" that can be studied to implement similar social structures in imbalanced regions.